Day 2: Machine Learning & Neural Networks

Complete High-Yield Reference Handbook & Study Notes

1. AI vs. Machine Learning vs. Deep Learning

• **Artificial Intelligence (AI):** Any system that mimics human decisions. Does NOT have to learn (can be 50 hard-coded if/else rules).
• **Machine Learning (ML):** Systems that discover patterns directly from past data without hard-coded rules.
• **Deep Learning (DL):** Machine learning powered by multi-layer Neural Networks. Automatically extracts features from raw images, audio, and text.

2. Learning Paradigms & Task Types

• **Supervised Learning:** Computer has an **Answer Key (y)**. Learns to predict future answers (e.g. house price, cancer detection).
• **Unsupervised Learning:** Computer has **NO Answer Key**. Finds natural clusters/patterns on its own (e.g. customer segmentation).
• **Regression:** Supervised task predicting a continuous **number** (e.g. Salary, Temperature, Used Car Price).
• **Classification:** Supervised task predicting a discrete **bucket/category** (e.g. Pass/Fail, Spam/Not Spam, Cat/Dog).

3. The 5 Core Algorithms Comparison Table

Algorithm	Type	Core Intuition	Best For
Linear Regression	Supervised (Regression)	Draws best-fit straight line: y = mx + b	Continuous numbers (Salaries, Prices)
Logistic Regression	Supervised (Classification)	S-curve estimating 0% to 100% probability	Yes/No decisions (Defaults, Churn)
Decision Tree	Supervised (Both)	Flowchart of Yes/No threshold questions	Easy business explanations
Random Forest	Supervised (Both)	Ensemble of 100+ Trees voting together	Tabular data, avoids overfitting
K-Means	Unsupervised (Clustering)	Groups unlabelled points into K center pins	Customer segmentation, groupings

4. Training, Testing, Overfitting & Metrics

Ensuring your model generalizes to unseen test data and grading its performance accurately.

Overfitting vs. Underfitting

• **Underfitting (Under-prepared):** Model is too simple. Performs poorly on both training and test sets.
• **Overfitting (Over-memorized):** Model memorized training noise. Scores 99% on training, drops to 55% on test exam! Fix with more data or Random Forests.
• **Regression Metrics: MAE** (average mistake size in real $), **RMSE** (penalizes massive mistakes), **R-Squared** (% variance explained, 0 to 1).
• **Classification Metrics: Accuracy** (% correct), **Precision** (few false alarms), **Recall** (catches every real positive case, vital in medicine), **F1** (harmonic balance).

5. Neural Networks & Activation Functions

How a Neuron Thinks: Multiplies inputs by weights and adds a bias: Sum = (X1*W1) + (X2*W2) + Bias.
Why Activation Functions? Without them, stacking 100 layers is still just a flat ruler. Activation functions add hinges so the network can bend to fit complex curves (cat faces, speech).
• **ReLU (Rectified Linear Unit):** If x < 0 -> 0; If x >= 0 -> x. Fast, acts like an automatic threshold switch (e.g., AC turns ON only above 24°C).
• **Sigmoid:** Squashes values between 0 and 1 (ideal for final Yes/No probability).
• **Softmax:** Converts multi-class outputs into percentages that sum to 100%.

6. Loss Function & Gradient Descent

The Foggy Mountain Analogy:
• **Loss Function:** The error scoreboard measuring how wrong the model is. Goal = reach 0 error at the valley bottom.
• **Gradient Descent:** Stepping downhill along the steepest slope to reduce loss step-by-step.
• **Learning Rate (Step Size):**
  • *Too Large:* Giant leap → overshoots the valley completely & loss jumps erratically!
  • *Too Small:* Tiny baby steps → takes years to reach the valley bottom.
• **Backpropagation:** Calculating the gradient backwards through the network to tweak every neuron's weight.